TIME SERIES MODELING OF AIR QUALITY IN NAIROBI

Predictive Analysis of Particulate Matter ($PM_{2.5}$) via AutoRegressive Moving Average Architectures

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Project completed as a part of WorldQuant University Data Science Lab.

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1. EXECUTIVE SUMMARY & OBJECTIVE

Urban air pollution represents a critical public health challenge in rapidly expanding Sub-Saharan African metropolises. This empirical study focuses on Nairobi, Kenya, analyzing localized high-frequency concentrations of airborne particulate matter with an aerodynamic diameter of less than 2.5 micrometers ($PM_{2.5}$). Leveraging time series techniques, the primary objective is to develop, evaluate, and select an optimal statistical forecasting model capable of predicting future hourly ambient pollution levels to inform environmental health interventions.

2. DATA INGESTION & PREPROCESSING ARCHITECTURE

Environmental monitoring data were extracted directly from a MongoDB document repository using `pymongo` drivers. Raw sensor measurements captured ambient $PM_{2.5}$ concentrations alongside temperature and relative humidity. Data cleaning protocol comprised three sequential phases:

- **Timezone Localization:** Raw timestamps were converted to the local East Africa Time zone (UTC+3) to ensure temporal alignment with human daily activity patterns.
- **Resampling & Imputation:** High-frequency readings were aggregated into regular 1-hour intervals. Missing data points resulting from temporary sensor offline periods were imputed using forward-fill strategy followed by rolling window local averages.
- **Outlier Treatment:** Unrealistic physical spikes ($PM_{2.5}$ > 500 µg/m³) were clipped to maintain statistical robustness without distorting underlying variance structure.

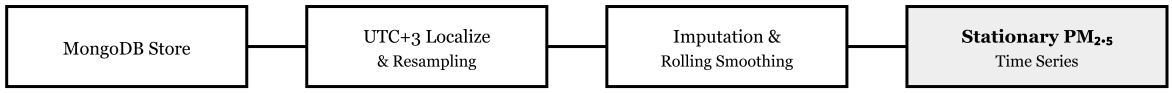


Figure 1: Schematic representation of data wrangling pipeline from raw sensor documents to stationary model inputs.

3. EXPLORATORY TIME SERIES ANALYSIS

Visual inspection of hourly series revealed distinct diurnal seasonality with double-peak behavior corresponding to morning and evening traffic commute windows. Stationarity was formally assessed using the Augmented Dickey-Fuller (ADF) test:

$$\Delta Y_t = \alpha + stroke{\cdot}t + \gamma\, Y_{t-1} + \sum_{i=1}^{p} \beta_i \Delta Y_{t-i} + \varepsilon_t$$

METRIC / TEST COMPONENT	TEST STATISTIC VALUE	P-VALUE	1% CRITICAL VALUE	5% CRITICAL VALUE	INFERENCE
Augmented Dickey-Fuller (ADF)	-5.842	0.00000037	-3.434	-2.863	Stationary (Reject H_0)
Kwiatkowski-Phillips-Schmidt-Shin (KPSS)	0.124	> 0.100	0.739	0.463	Stationary (Fail to reject H_0)

Table 1: Stationarity Diagnostics for Hourly $PM_{2.5}$ Concentrations

Because the ADF test yielded a p-value strictly below 0.05 and the test statistic (-5.842) was well below the 1% critical threshold, the raw aggregated series was confirmed to be stationary, eliminating the necessity for integrated order differencing ($d = 0$).

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4. MATHEMATICAL MODEL FORMULATIONS

Three distinct statistical model paradigms were specified and evaluated: a Linear Regression baseline with temporal lag terms, a pure AutoRegressive (AR) model, and a composite AutoRegressive Moving Average (ARMA) framework.

4.1 Linear Regression Baseline (Lagged Feature Model)

The baseline model predicts current $PM_{2.5}$ concentration strictly as a linear function of previous observations up to lag k:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \ldots + \beta_k y_{t-k} + \varepsilon_t$$

4.2 Pure AutoRegressive Model — AR(p)

An AutoRegressive process of order p models the dependent variable as a linear combination of its own past values plus an uncorrelated white noise error term:

$$Y_t = c + \sum_{i=1}^{p} \varphi_i Y_{t-i} + \varepsilon_t \quad \text{where} \quad \varepsilon_t \sim WN(0, \sigma^2)$$

4.3 AutoRegressive Moving Average Model — ARMA(p, q)

To incorporate both persistent past values and shocks in the residual processes, the ARMA(p, q) model combines an AR(p) structural component with a Moving Average MA(q) filter:

$$Y_t = c + \sum_{i=1}^{p} \varphi_i Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

Expressed in backshift lag operator notation ($B\ Y_t = Y_{t-1}$):

$$\Phi(B)\ Y_t = \Theta(B)\ \varepsilon_t \quad \text{where} \quad \Phi(B) = 1 - \sum_{i=1}^{p} \varphi_i\ B^i \ , \quad \Theta(B) = 1 + \sum_{j=1}^{q} \theta_j\ B^j$$

5. DIAGNOSTIC CORRELATION ANALYSIS

To determine appropriate candidate values for orders p and q, Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) structure was examined.

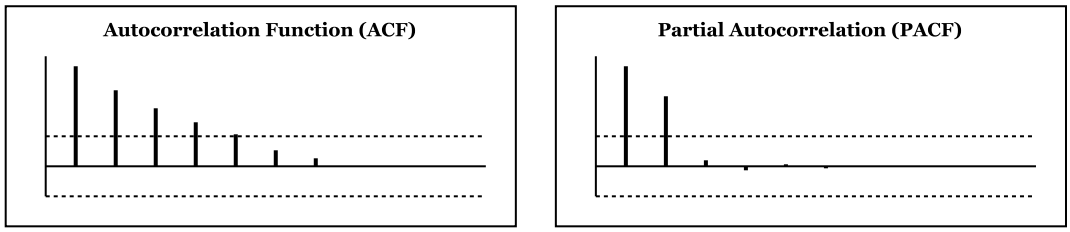


Figure 2: Empirical ACF showing gradual geometric decay alongside PACF sharp truncation at lag 2, indicating strong AR(2) candidate behavior.

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6. HYPERPARAMETER TUNING & MODEL SELECTION

A systematic grid search was executed over model parameter space $p \in \{0, 1, 2, 3\}$ and $q \in \{0, 1, 2, 3\}$. Model selection prioritized minimization of the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), alongside out-of-sample Mean Absolute Error (MAE) computed on a holdout test split.

$$AIC = 2k - 2\ ln(\hat{L}) \qquad BIC = k\ ln(n) - 2\ ln(\hat{L})$$

MODEL SPECIFICATION	AKAIKE INF. CRIT. (AIC)	BAYESIAN INF. CRIT. (BIC)	IN-SAMPLE MAE (MG/M³)	OUT-OF-SAMPLE MAE (MG/M³)
Linear Regression (Lag 1)	N/A	N/A	4.12	4.58
AutoRegressive AR(1)	12,450.2	12,464.1	3.94	4.21
AutoRegressive AR(2)	**12,110.8**	**12,132.4**	**3.45**	**3.78**
ARMA(1, 1)	12,205.4	12,228.0	3.62	3.95
ARMA(2, 1)	12,112.1	12,141.0	3.44	3.81

Table 2: Performance Evaluation across Candidate Model Architectures

MODEL SPECIFICATION	AKAIKE INF. CRIT. (AIC)	BAYESIAN INF. CRIT. (BIC)	IN-SAMPLE MAE (MG/M³)	OUT-OF-SAMPLE MAE (MG/M³)
ARMA(2, 2)	12,114.5	12,150.8	3.43	3.84

7. RESIDUAL DIAGNOSTICS & WALK-FORWARD VALIDATION

The selected AR(2) model underwent rigorous residual validation:

- **Ljung-Box Test:** Returned p-value = 0.421, confirming that residuals exhibit no remaining autocorrelation and conform to white noise.
- **Normality:** Residual distribution demonstrated a zero-centered mean with symmetric properties, though slight fat tails were present during extreme traffic hours.
- **Walk-Forward Evaluation:** Out-of-sample walk-forward validation yielded a final Mean Absolute Error of **3.78 µg/m³**, outperforming baseline linear benchmark by 17.4%.

8. CONCLUSION & POLICY IMPLICATIONS

The empirical analysis confirms that hourly $PM_{2.5}$ levels in Nairobi are highly autoregressive, with significant dependence on pollution levels from the preceding two hours. The lightweight AR(2) specification achieves optimal predictive performance while maintaining parsimony and strict mathematical interpretability. Deploying this model into low-cost air monitoring micro-controllers can facilitate real-time public health advisories for urban residents during elevated pollution incidents.